

IMASHA DEWMI

FULL-STACK DEVELOPER INTERN

- No. 201/8 3rd Lane , Pallimulla Panadura •+94 77 577 1003 • imashadewmi557@gmail.com
- Linkedin:** <https://www.linkedin.com/in/imasha-dewmi-a464212b9>
- GitHub:** <https://github.com/Dewmi2004>

SUMMARY

Motivated and self-driven Full-Stack Developer with a solid foundation in designing and building scalable backend systems, RESTful APIs, and web applications, seeking to gain real-world industry experience. Skilled in Node.js, TypeScript, Express.js, and React, with a strong understanding of SQL/NoSQL databases. Detail-oriented and analytical, with practical knowledge in Artificial Intelligence and Machine Learning, including data preprocessing, model evaluation, and supervised learning using Python (NumPy, Pandas, Scikit-learn). Adaptable and self-driven, with a passion for building intelligent, user-centric applications through data-driven approaches, clean and efficient code, and continuous learning of modern technologies.

SKILLS

TECHNICAL SKILLS

- **Backend:** Node.js, TypeScript, Express.js, REST APIs, Microservices
- **Frontend:** React.js, React Native, HTML5, CSS3, Tailwind CSS, Bootstrap
- **Databases:** MySQL, MongoDB, SQL Server, Firebase
- **Tools:** Git, GitHub, Docker, Postman, Figma, Hibernate, Google Colab , Kaggle
- **Languages:** TypeScript, JavaScript, Java, Python, C++
- **AI/ML:** Supervised Learning , Unsupervised Learning , Deep Learning , Neural Network , CNN , LLM , RAG Systems , Ollama , Bias Mitigation , Data Preprocessing

SOFT SKILLS

- Teamwork
- Problem Solving
- Leadership
- Critical Thinking

LANGUAGES

- **Sinhala** - Native
- **English** - Advanced
- **Tamil** - Intermediate

EDUCATION

Institute of Software Engineering (IJSE)

Aug 2024 - Present

- BSc (Hons) in Software Engineering

Sri Sumangala Girl's School Panadura (A/L)

2023 (Jan 2024)

- Completed G.C.E. Advanced Level in Science Stream (Biology , Chemistry , Physics)

CERTIFICATIONS

- **Certified AI & Machine Learning Engineering (CAME)** – Institute of Software Engineering (In Progress)
- **Cisco IT Essentials** – Cisco Networking Academy (2025)
- **Java Fundamental to Intermediate Course** – Academy of Future Robotics (2025)
- **C++ Fundamental to Intermediate Course** – Academy of Future Robotics (2025)

PROJECTS

DewCode – AI-Powered Developer Collaboration Platform

Full-stack browser-based IDE combining AI-assisted development, secure Docker execution, real-time collaboration, and subscription-based features.

- Built a browser-based IDE with Monaco Editor, project management, integrated terminal, and multi-language support.
- Implemented AI code generation, suggestions, corrections, debugging, and code review using locally hosted Ollama models.
- Developed Docker-based sandbox execution with isolated containers and resource limits for secure code execution.
- Implemented JWT authentication, Developer role, and Free/Premium subscription plans with PayHere payment integration.
- Added Premium real-time collaboration with multi-user editing, live cursor tracking, Socket.IO synchronization, and individual/group chat.

Tech: React.js, TypeScript, Node.js, Express.js, MongoDB, Socket.IO, Ollama, Docker, GitHub API, JWT, PayHere

GitHub: <https://github.com/Dewmi2004/DewCode>

Smart Event Ticketing & Reservation System

Full-stack event management and ticketing system built with Spring Boot, featuring secure reservations, real-time seat locking, PayHere payments, digital QR tickets, notifications, coupons, and refund management.

- Developed secure event booking and temporary seat locking to prevent double bookings and manage real-time seat availability.
- Integrated PayHere payment gateway for online ticket payments with payment status handling.
- Implemented mock refund processing, including refund requests, status tracking, and refund records.
- Developed QR code-based e-tickets and custom QR scanning for ticket identification and validation.
- Implemented coupon/discount management and booking, payment, event, and customized notifications using Email and Text.Ik SMS.
- Implemented JWT-based Admin/User role access control and an admin dashboard for managing events, bookings, payments, refunds, coupons, and analytics.

Tech: Spring Boot, Spring Security, JWT, Spring Data JPA, MySQL, HTML, CSS, JavaScript, PayHere, Text.Ik, SMTP, QR Code, Maven

GitHub: https://github.com/Dewmi2004/Smart_Event_Ticketing_System

NovaX — Credit Card Default Risk Analysis | Team Leader

- Led a 5-member data science project analyzing 30,000 credit card customer records to identify default-risk patterns and support data-driven lending decisions.
- Developed the Business Strategy & API Integration module, integrating external economic indicators through REST APIs and translating risk segments into actionable lending strategies.
- Defined segment-based recommendations for Healthy, Watchlist, At-Risk, and Critical customers, including approval conditions, monitoring, higher interest rates, collateral requirements, or rejection.
- Contributed to project methodology, business insights, documentation, and final presentation while coordinating team deliverables.

Tech: Python, Pandas, NumPy, Requests, REST APIs, SciPy, Matplotlib, Seaborn, Jupyter Notebook

GitHub: <https://github.com/Dewmi2004/NovaX--Group-project-of-Python-for-DS-AI>

FairVision — CNN-Based Age Classification & Fairness Analysis

AI/ML project developing a 9-class age group classification system using a custom CNN trained from scratch on the FairFace dataset, with demographic fairness evaluation and bias mitigation across race and gender groups.

- Developed and trained a VGG-inspired CNN from scratch for age group classification across nine age categories, achieving **58.07%** test accuracy.
- Evaluated model performance and demographic disparities using accuracy, F1-score, race gaps, and gender gaps across seven race and two gender groups.
- Implemented Focal Loss, soft class weighting, and WeightedRandomSampler to address class imbalance and mitigate demographic performance gaps.
- Reduced the race performance gap by 36% using Focal Loss with soft class weights while maintaining competitive overall accuracy.
- Deployed the fairness-mitigated model as a Streamlit web application with image upload, Top-3 predictions, confidence scores, and responsible-use limitations.

Tech: Python, PyTorch, CNN, FairFace, NumPy, Pandas, Scikit-learn, Matplotlib, Streamlit

GitHub: https://github.com/Dewmi2004/FairVision-CNN_Age_Group_Classification_Fairness_Audit

Link: <https://fairvision-cnnagegroupclassificationfairnessaudit-dzdo3brxbvn6.streamlit.app>

EstateGo – Real Estate Property Listing Mobile App

Cross-platform React Native real-estate application for browsing, managing, and listing properties with favorites, payments, maps, notifications, and multi-language support.

- Developed property listing, search, filtering, CRUD, favorites, and image upload features.
- Integrated PayHere payment gateway with a simulated payment flow and optional sandbox checkout for property listing payments.
- Implemented rule-based EstateBot, Google Maps directions, local notifications, and English/Sinhala/Tamil language support.
- Built a responsive iOS, Android, and Web application using a reusable TypeScript component architecture.

Tech: React Native, Expo, TypeScript, Redux Toolkit, React Navigation, Axios, AsyncStorage, Express.js

GitHub: <https://github.com/Dewmi2004/EstateGo>

REFERENCES

Udara San

Senior Lecturer @ IJSE

Email: udara@ijse.lk

Prasad Waduge

CEO @ IJSE

Email: prasad@ijse.lk